

ADAM PARKOSIDIS

Science Park 904, 1098XH
Anton Pannekoek Institute for Astronomy
University of Amsterdam
Amsterdam, North Holland, The Netherlands

EDUCATION

Master of Science <i>Physics and Astronomy</i> 'Cum Laude'	2021 – 2023
University of Amsterdam-Vrije Universiteit Amsterdam, The Netherlands	
• Thesis: <i>Mass Transfer in Hierarchical Triple Systems: The Case of Roche-Lobe filling Outer Star</i>	
Bachelor of Science <i>Physics</i>	2013 – 2020
Aristotle University of Thessaloniki, Greece	
• Thesis: <i>Dynamic Perturbations in Protoplanetary Cosmic Dust Discs in the Presence of a Giant Planet: Possible Detection with ALMA</i>	

PUBLICATIONS

Reframing the wide eccentric binary problem: Eccentricity as a probe of mass-transfer physics	2026, accepted ApJ
A. Parkosidis, S. Toonen, E. Laplace, V. Schaffenroth	
Rethinking mass transfer: a unified semianalytical framework for circular and eccentric binaries. II. Orbital evolution due to non-conservative mass transfer	2026, A&A
A. Parkosidis, S. Toonen, E. Laplace, F. Dosopoulou	
Rethinking mass transfer: a unified semianalytical framework for circular and eccentric binaries. I. Orbital evolution due to conservative mass transfer	2026, A&A
A. Parkosidis, S. Toonen, F. Dosopoulou, E. Laplace	
Episodic outbursts during brown dwarf formation	2025, A&A
A. Parkosidis, D. Stamatellos, B. Riaz	

SCIENTIFIC EXPERIENCE

PhD Candidate in Astrophysics	2023 – Present
Anton Pannekoek Institute, UvA, The Netherlands	
Technical Support Scientist	2023
ASTRON, Dwingeloo, The Netherlands	
• Supported LOFAR Unified Calculator for Imaging (LUCI) tool development by testing, debugging, and integrating new Python functionalities	
• Developed observation and processing templates utilizing LOFAR's Telescope Manager Specification System following proposal approval	
• Authored documentation and user guides on instrument capabilities and operational procedures	
• Performed quality control of data products generated by the Long-Term Archive (LTA) pipelines, producing reports and performance summaries	
Research Intern in Computational Astrophysics	2021
Jeremiah Horrocks Institute, UCLan, United Kingdom	
• Applied the PHANTOM smoothed particle hydrodynamics (SPH) code for large-scale numerical simulations of protoplanetary disks	
• Utilized the UCLan High-Performance Computing (HPC) cluster for parallel execution and resource-optimized SPH model runs	
• Processed, analyzed, and visualized simulation data using Python (NumPy, pandas, matplotlib, seaborn) and SPLASH	
Space Debris Data Analyst	2020 – 2023
SpaceDot, Thessaloniki, Greece	
• Collaborated with 80+ engineers and scientists to integrate debris mitigation data into mission design	

- Performed orbit propagation and collision analyses with ESA's DRAMA software to ensure compliance with international space debris mitigation standards
- Developed Python scripts for automating orbital simulations, data analysis, and visualization of on-orbit collision risk scenarios
- Contributed technical content to the System Design and Mission Requirements (SDMR) and Orbital Collision Risk Assessment (OCRA) documents for the Critical Design Review (CDR) phase

FUNDING & GRANTS

Leids Kerkhoven-Bosscha Fonds (LKBF)

Years granted: 2024, 2025, 2026

TALKS

<i>When Mass Transfer Makes Orbits Eccentric: Connecting Theory and Observation</i>	2026
European Astronomical Society	Lausanne, CHE
<i>Eccentricity as a Probe of Mass-transfer Physics</i>	2026
Colloquium: Institute of Astronomy	KU Leuven, BEL
<i>Mass Transfer in Eccentric Binaries: A Unified Framework & New Insights</i>	2025
Binary Stars in a New Era	Lijiang, CHN
<i>Mass Transfer in Eccentric Binaries: A Unified Framework & New Insights</i>	2025
Netherlands Astronomy Conference	Nijmegen, NL

TECHNICAL SKILLS

Operation Systems: Linux (Ubuntu), Microsoft Windows
Programming: Python, Bash, C, F77/F95
Machine Learning & Deep Learning: PySR, Keras, Pandas, PyTorch, Scikit-Learn, TensorFlow
Data Visualization: Matplotlib, Seaborn, Plotly, gnuplot
Document Creation: Microsoft Office Suite, LibreOffice Suite, Google Suite, LaTeX
Other Technologies: Smoothed Particle Hydrodynamics (SPH), Git

SCIENTIFIC COMPUTING

SPH codes: PHANTOM, SEREN
Stellar Evolution codes: MESA
N-body codes: SWIFT, HUAYNO
Python Packages: NumPy, SciPy, SymPy, DISKLAB, Astropy, PICASO
Space Debris Risk Assessment & Mitigation Analysis: DRAMA
Other software: AMUSE, Mathematica

TEACHING EXPERIENCE & STUDENT SUPERVISION

University Lectures	2026
• 'Convection' (M.Sc. course: 'Structure & Evolution of Stars')	
Summer projects	2024
• Magdalena Andrea Vilaxa Campos "Modeling Mass Transfer Between Stars" (MSc level)	
Teaching Assistant	2022 – 2025
Anton Pannekoek Institute, UvA, The Netherlands	
• 'Atmospheres & Radiative Transfer' (M.Sc. course, 4 times)	
• 'Structure & Evolution of Stars' (M.Sc. course, 2 times)	
• 'Disks & Accretion' (M.Sc. course)	
AUTh, Greece	2018
• 'Observational Astronomy' (BSc course)	
Private Tutor	2018 – 2019
Thessaloniki, Greece	
• Tutored and prepared secondary education students for their final exams in the school subjects of Physics and Math	

TRAININGS & SCHOOLS

<i>Binary compact objects on eccentric orbits and their gravitational-wave signatures</i> Corfu, Greece	2026
<i>ICESUN Summer School 2025: Binary Stars and Compact Objects</i> Yunnan Observatories, China	2025
<i>MESA Summer School</i> KU Leuven, Belgium	2025
<i>Nordic Winter School on Multimessenger Astrophysics</i> Skeikampen, Norway	2024
<i>Mastering your PhD</i> University of Amsterdam, The Netherlands	2024

COLLABORATION MEMBERSHIPS

The Binariness at Low Metallicity (BLOeM)	2026 – Present
Laser Interferometer Space Antenna (LISA)	2025 – Present

LANGUAGES

Greek: Native
English: Fluent
Dutch: Elementary
German: Elementary

VOLUNTEERISM

Guide and telescope operator Observatory of AUTH	2018 – 2019
Operated the optical telescope at the Aristotle University of Thessaloniki Observatory during visits by primary, junior high, and high school students.	